Energi Potensial Listrik dan Potensial Listrik

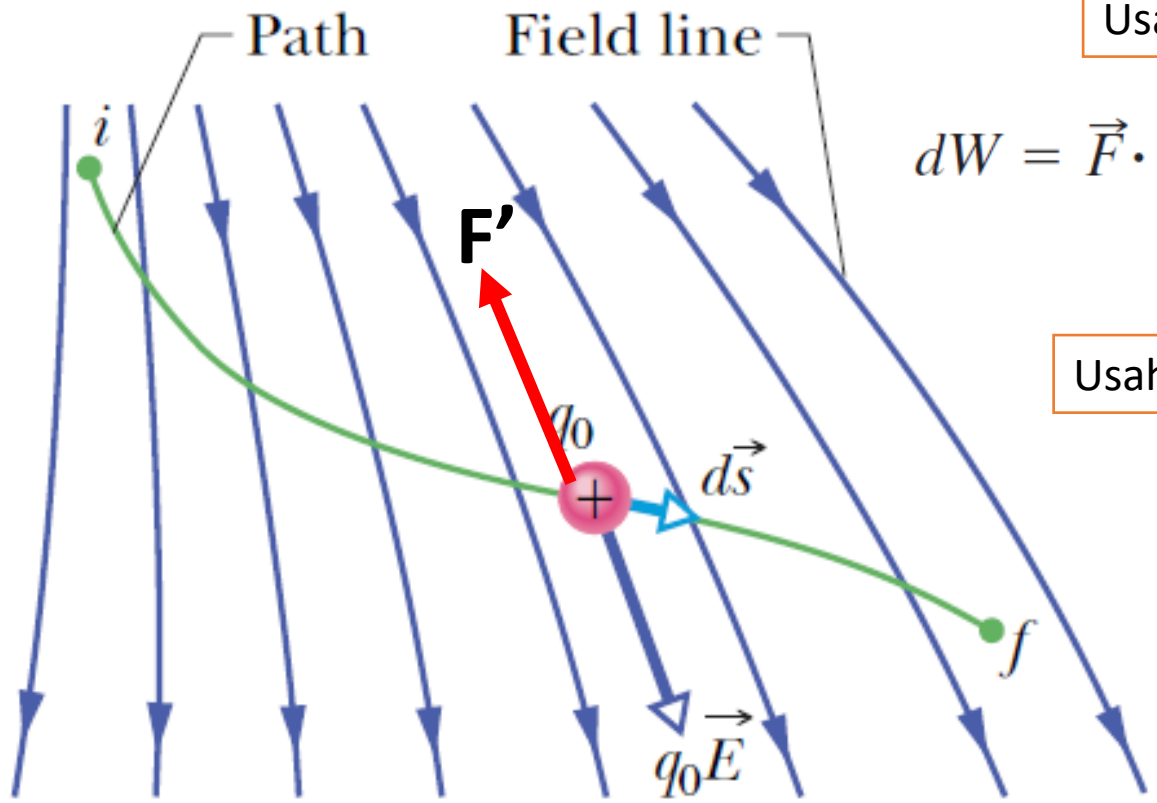


Defrianto Pratama, S.Pd., M.Si

Sumber Bacaan : Halliday ch. 24

- **ENERGI POTENSIAL LISTRIK**

Usaha yang diperlukan untuk membawa muatan Q melalui medan listrik **E** dari suatu titik ke titik lain :



Usaha yang dilakukan oleh gaya listrik

$$dW = \vec{F} \cdot d\vec{s}.$$

$$W = q_0 \int_i^f \vec{E} \cdot d\vec{s}.$$

Usaha yang dilakukan oleh gaya F'

$$W' = -W = -q_0 \int_i^f \vec{E}.\overrightarrow{ds}$$

Beda Potensial Listrik

Energi potensial listrik per satuan coloumb

$$\Delta V = V_f - V_i = -W/q_0$$

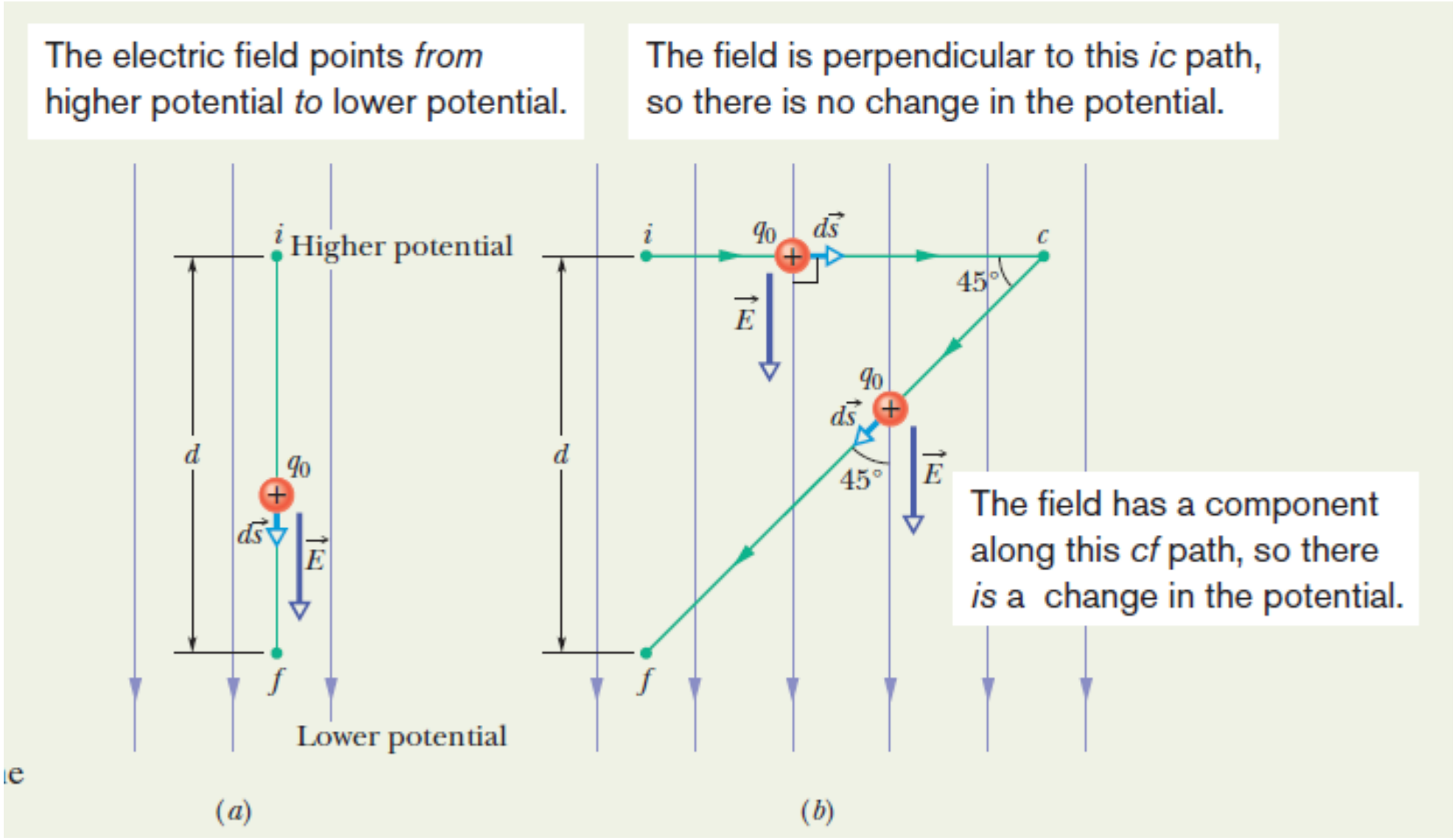
Satuan beda potensial adalah Joule/Coloumb atau Volt

$$V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s}.$$

If we set potential $V_i = 0$ (di posisi tak hingga dari sumber medan)

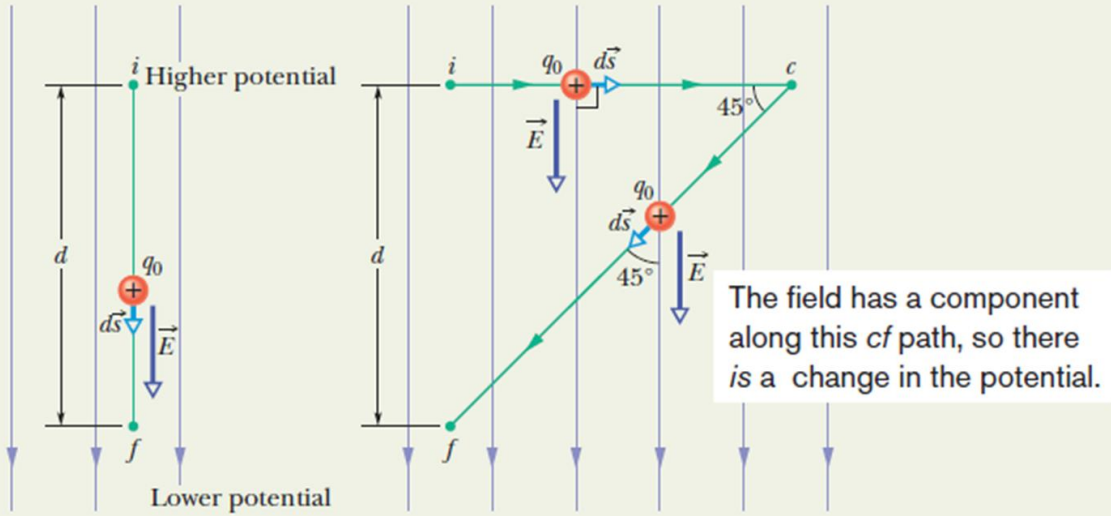
$$V = -\int_i^f \vec{E} \cdot d\vec{s}.$$

Contoh : Tentukan beda potensial antara titik i dan f untuk dua kasus berikut !



The electric field points *from* higher potential *to* lower potential.

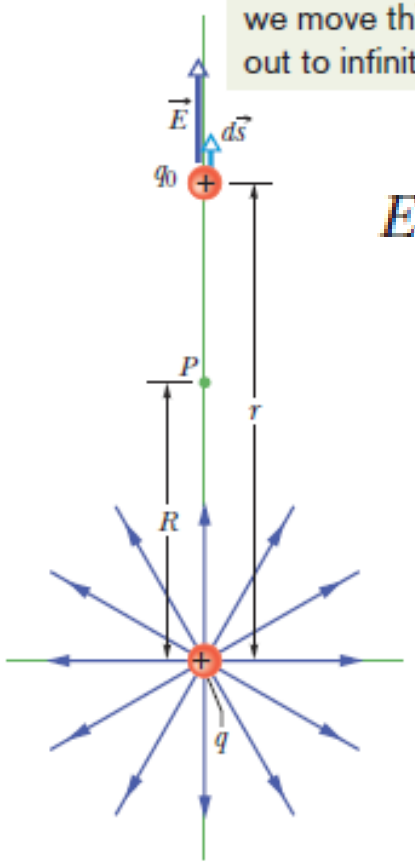
The field is perpendicular to this *ic* path, so there is no change in the potential.



The field has a component along this *cf* path, so there *is* a change in the potential.

Potensial pada muatan titik

To find the potential of the charged particle, we move this test charge out to infinity.



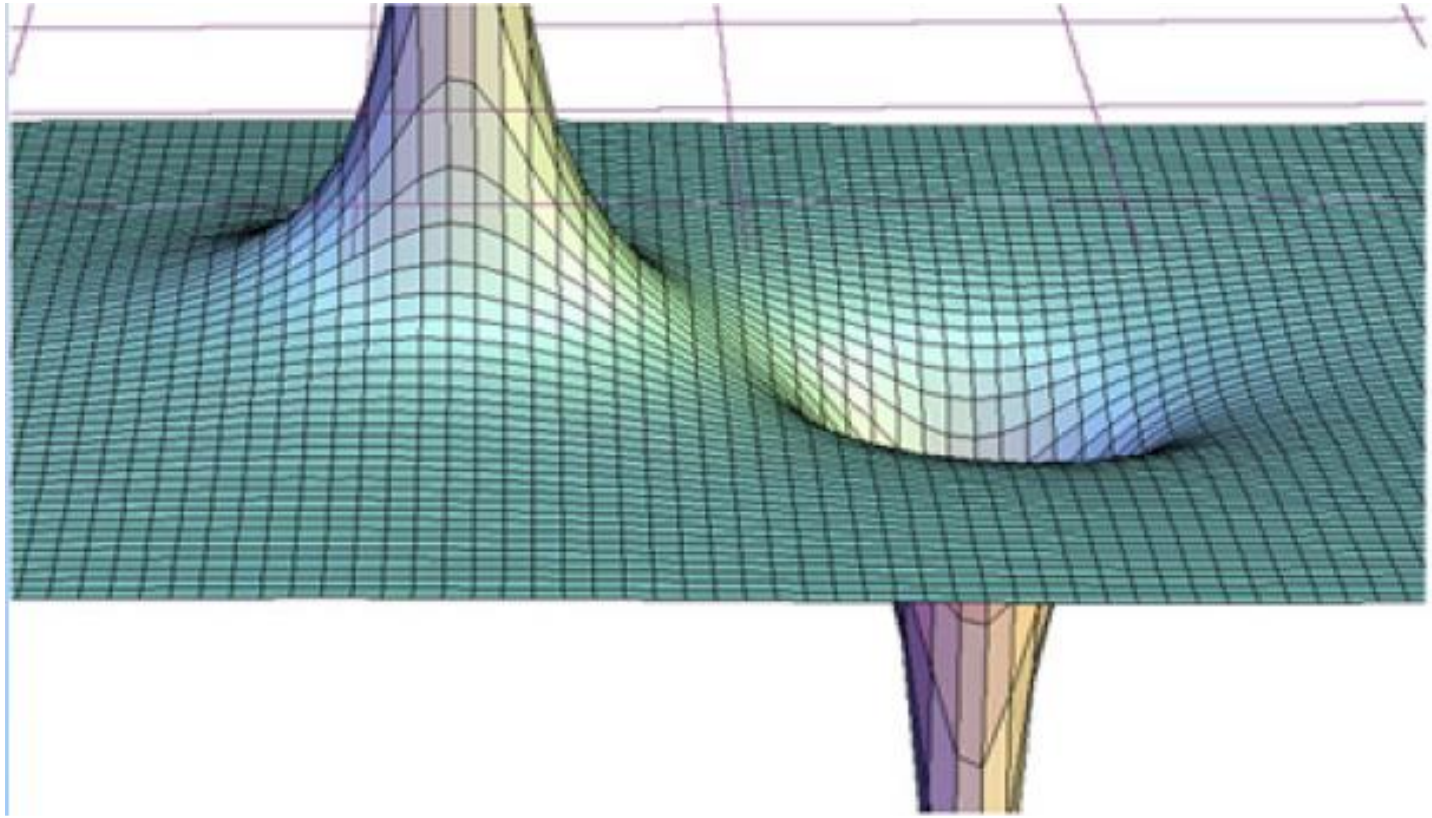
$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}.$$

$$V_f - V_i = -\int_i^f \vec{E}\cdot d\vec{s}.$$

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$

Bentangan Potensial

Muatan Positif



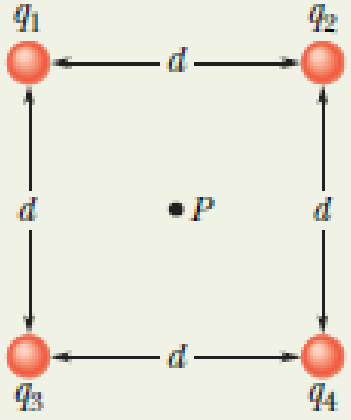
Muatan Negatif

Potensial pada distribusi Muatan Titik

Potensial adalah besaran sekalar maka total beda potensial adalah penjumlahan biasa dari beda potensial yang dihasilkan setiap muatan

$$V = \sum_{i=1}^{n} V_i = \frac{1}{4\pi\varepsilon_0} \sum_{i=1}^{n} \frac{q_i}{r_i} \qquad (n \text{ point charges}).$$

Contoh :



$q_1 = +12$ nC, $q_3 = +31$ nC,
$q_2 = -24$ nC, $q_4 = +17$ nC.
d is 1.3 m

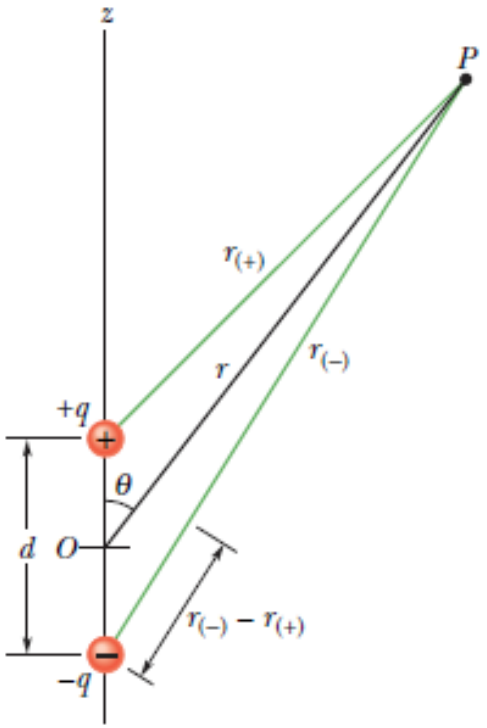
$$V = \sum_{i=1}^{4} V_i = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{r} + \frac{q_2}{r} + \frac{q_3}{r} + \frac{q_4}{r}\right).$$

The distance r is $d/\sqrt{2}$, which is 0.919 m, and the sum of the charges is

$$q_1 + q_2 + q_3 + q_4 = (12 - 24 + 31 + 17) \times 10^{-9}\,\text{C} = 36 \times 10^{-9}\,\text{C}.$$

Thus,
$$V = \frac{(8.99 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(36 \times 10^{-9}\,\text{C})}{0.919\,\text{m}} \approx 350\,\text{V}. \qquad \text{(Answer)}$$

Potensial listrik dipol

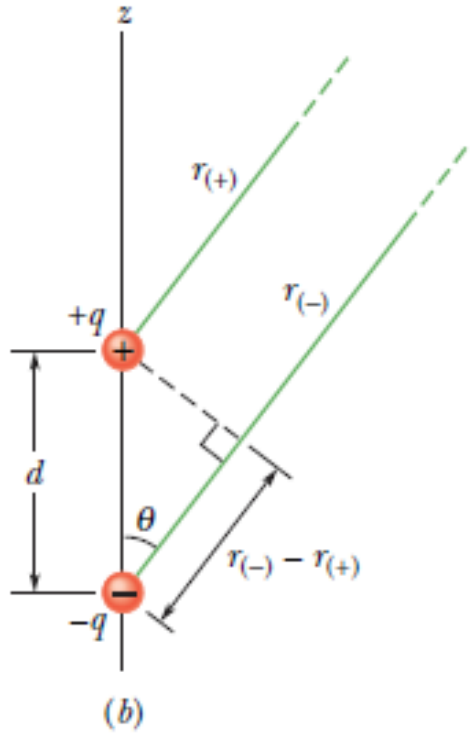


$$V = \sum_{i=1}^{2} V_i = V_{(+)} + V_{(-)} = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r_{(+)}} + \frac{-q}{r_{(-)}}\right)$$

$$= \frac{q}{4\pi\varepsilon_0}\frac{r_{(-)} - r_{(+)}}{r_{(-)}r_{(+)}}.$$

$$r \gg d,$$

$$r_{(-)} - r_{(+)} \approx d\cos\theta \quad \text{and} \quad r_{(-)}r_{(+)} \approx r^2.$$



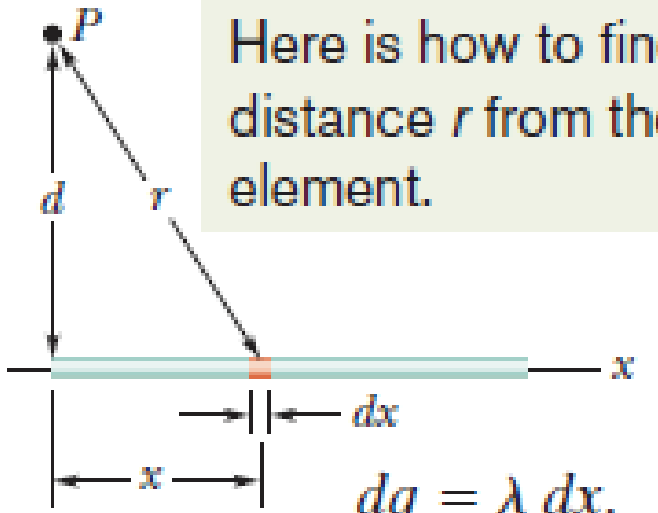
$$V = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2} \qquad \text{(electric dipole)},$$

Potensial pada distribusi Muatan kontinyu

$$dV = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r} \quad \Rightarrow \quad V = \int dV = \frac{1}{4\pi\varepsilon_0}\int \frac{dq}{r}.$$

Contoh : tentukan potensial dititik P

$$dV = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda\, dx}{(x^2 + d^2)^{1/2}}.$$



Here is how to find distance *r* from the element.

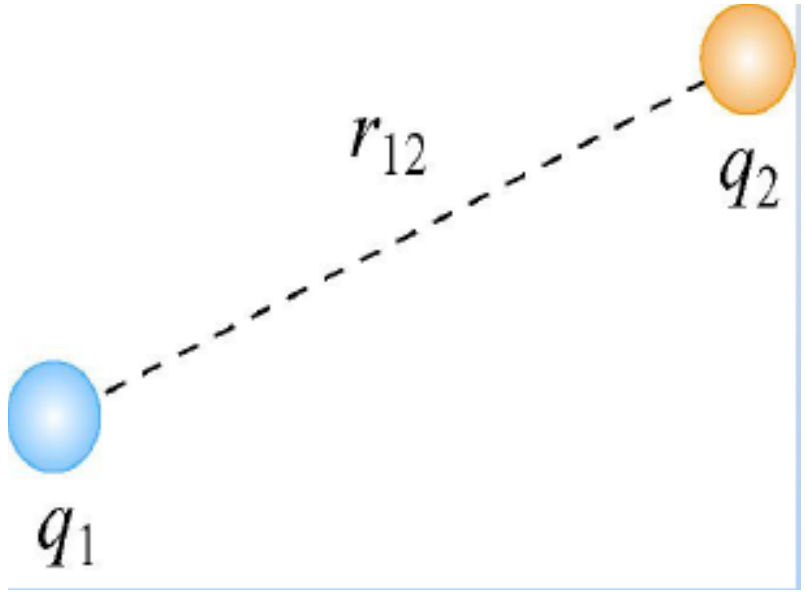
$$dq = \lambda\, dx.$$

$$V = \frac{\lambda}{4\pi\varepsilon_0}\ln\left[\frac{L + (L^2 + d^2)^{1/2}}{d}\right].$$

Energi Konfigurasi

Berapa besar energi untuk menyimpan dua muatan seperti pada gambar berikut?

1) Muatan pertama bebas
2) Muatan kedua melihat muatan pertama:

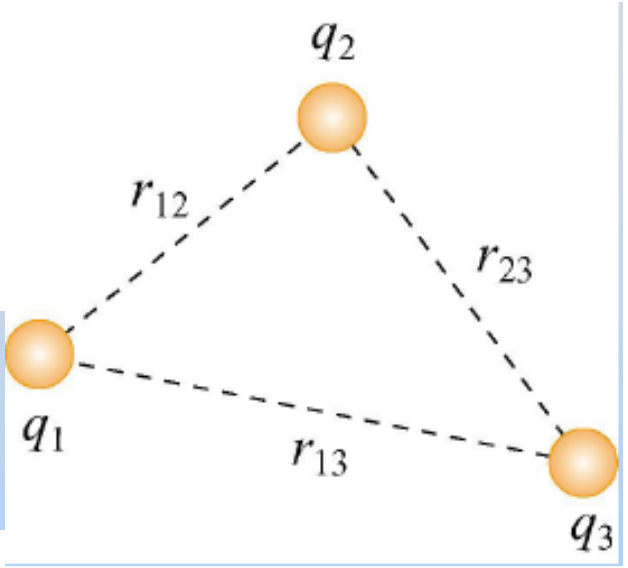


$$U_{12} = W_2 = q_2 V_1 = \frac{1}{4\pi\varepsilon_o}\frac{q_1 q_2}{r_{12}}$$

Energi Konfigurasi

Berapa besar energi untuk menyimpan tiga muatan seperti pada gambar berikut?

1) Sudah tahu dua muatan awal
2) Membawa muatan yang ketiga:



$$W_3 = q_3\left(V_1 + V_2\right) = \frac{q_3}{4\pi\varepsilon_0}\left(\frac{q_1}{r_{13}} + \frac{q_2}{r_{23}}\right)$$

Total energi konfigurasi :

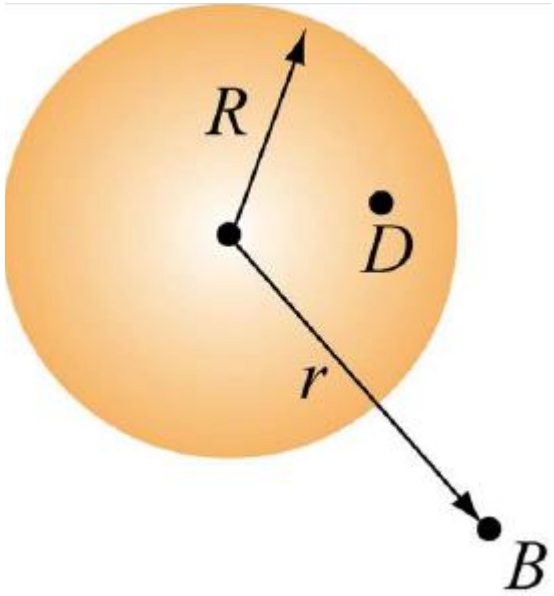
$$U = W_2 + W_3 = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}}\right) = U_{12} + U_{13} + U_{23}$$

- **Mencari V dari E**

Potensial dari Bola Padat Uniform Non-Konduktor

Dari Hukum Gauss

$$\vec{\mathbf{E}} = \begin{cases} \dfrac{Q}{4\pi\varepsilon_0 r^2}\hat{\mathbf{r}}, & r > R \\ \dfrac{Qr}{4\pi\varepsilon_0 R^3}\hat{\mathbf{r}}, & r < R \end{cases}$$



Gunakan

$$V_B - V_A = -\int_A^B \vec{\mathbf{E}} \cdot d\,\vec{\mathbf{s}}$$

Muatan Titik!

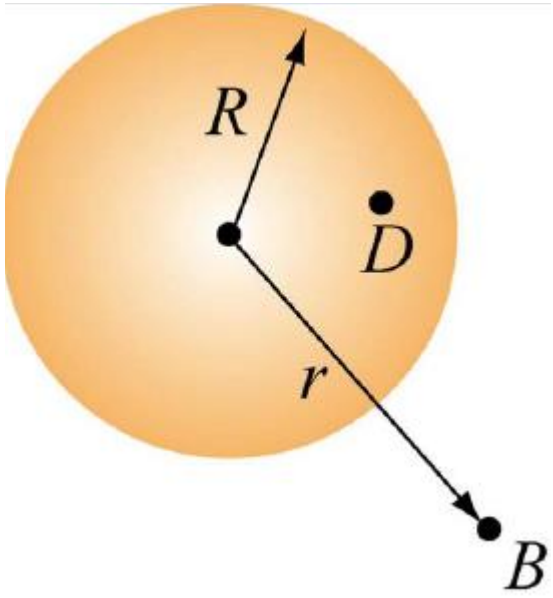
Daerah 1: $r > R$

$$V_B - \underbrace{V(\infty)}_{=0} = -\int_\infty^r \frac{Q}{4\pi\varepsilon_0 r^2} dr = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r}$$

Potensial dari Bola Padat Uniform Non-Konduktor

Dari Hukum Gauss

$$\vec{\mathbf{E}} = \begin{cases} \dfrac{Q}{4\pi\varepsilon_0 r^2}\hat{\mathbf{r}}, & r > R \\ \dfrac{Qr}{4\pi\varepsilon_0 R^3}\hat{\mathbf{r}}, & r < R \end{cases}$$



Gunakan

$$V_B - V_A = -\int_A^B \vec{\mathbf{E}} \cdot d\vec{\mathbf{s}}$$

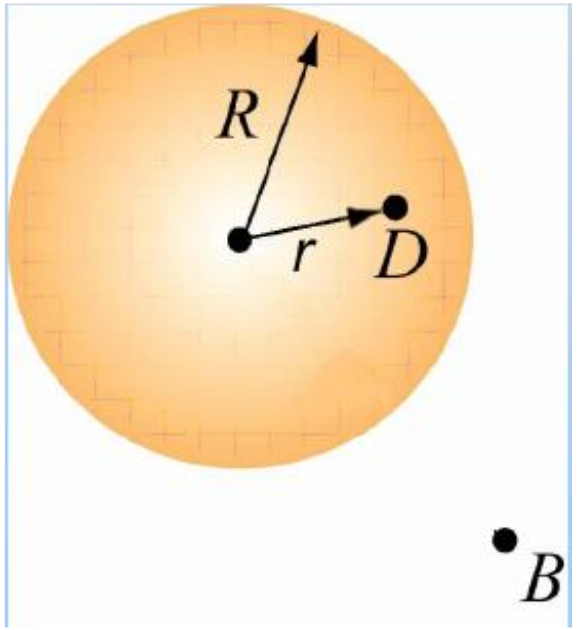
Muatan Titik!

Daerah 1: $r > R$

Potensial dari Bola Padat Uniform Non-Konduktor

Daerah 2: $r < R$

$$V_D - \underbrace{V(\infty)}_{=0} = -\int_{\infty}^{R} dr E(r > R) - \int_{R}^{r} dr E(r < R)$$



Potensial dari Bola Padat Uniform Non-Konduktor

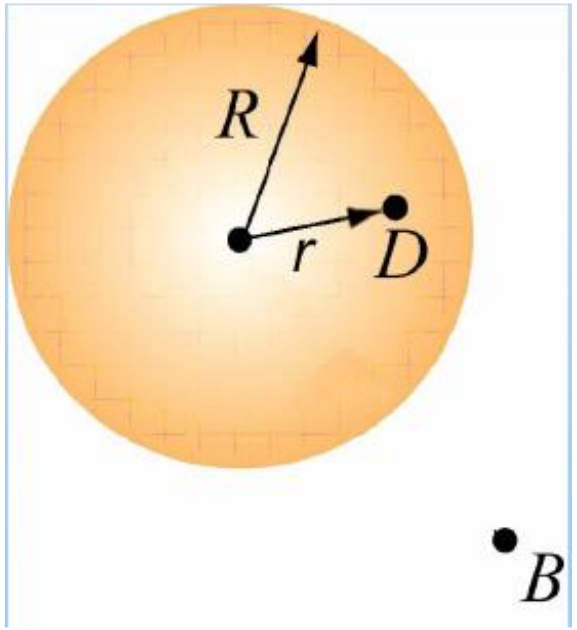
Daerah 2: $r < R$

$$V_D - \underbrace{V(\infty)}_{=0} = -\int_{\infty}^{R} dr E(r>R) - \int_{R}^{r} dr E(r<R)$$

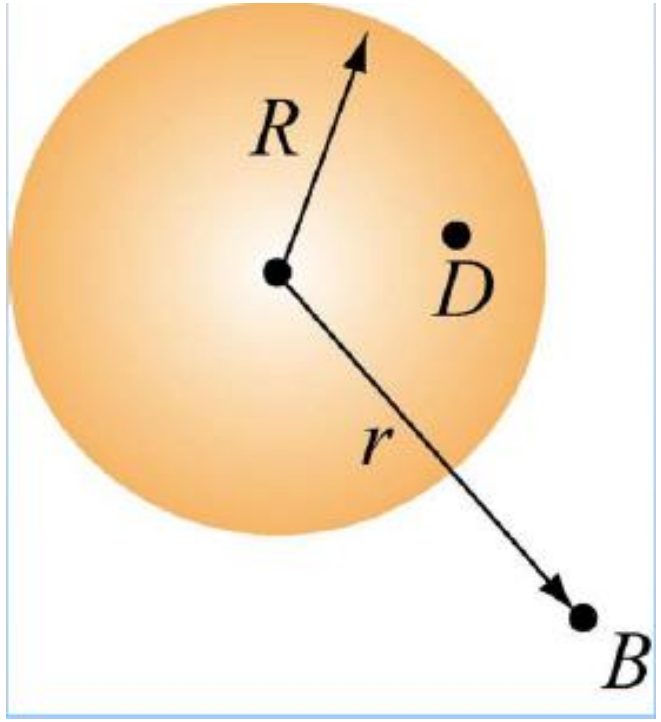
$$= -\int_{\infty}^{R} dr \frac{Q}{4\pi\varepsilon_0 r^2} - \int_{R}^{r} dr \frac{Qr}{4\pi\varepsilon_0 R^3}$$

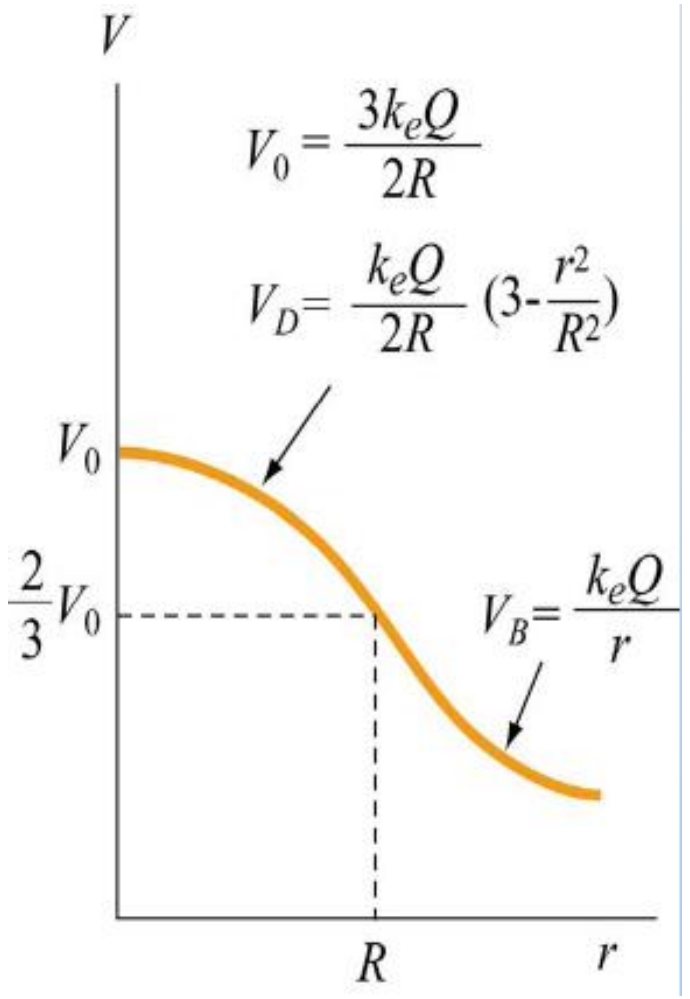
$$= \frac{1}{4\pi\varepsilon_0}\frac{Q}{R} - \frac{1}{4\pi\varepsilon_0}\frac{Q}{R^3}\frac{1}{2}\left(r^2 - R^2\right)$$

$$= \frac{1}{8\pi\varepsilon_0}\frac{Q}{R}\left(3 - \frac{r^2}{R^2}\right)$$



Potensial dari Bola Padat Uniform Non-Konduktor

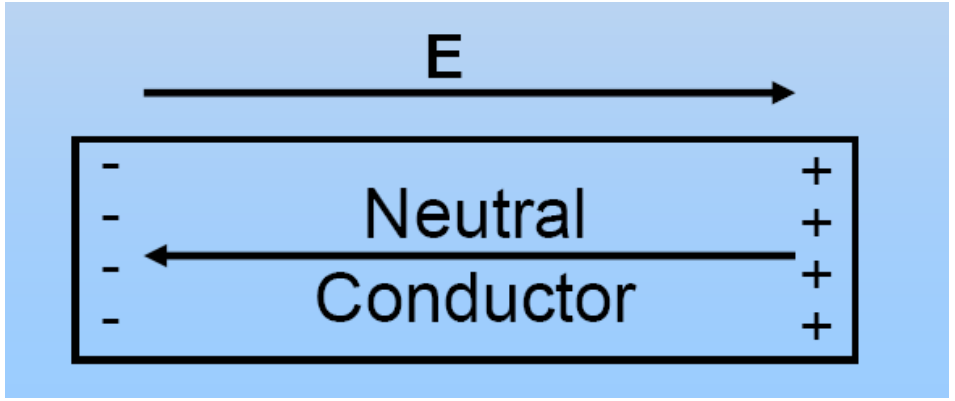




Konduktor

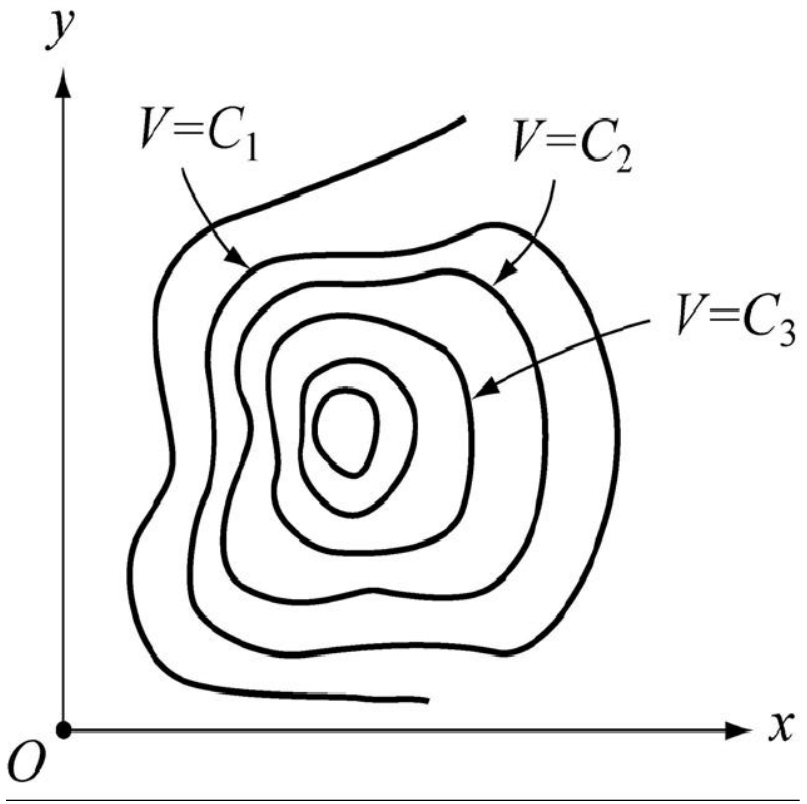
Konduktor memiliki muatan bebas

- E bernilai nol di dalam Konduktor
- Konduktor adalah objek equipotensial



$\mathbf{E}_{Total} = 0$

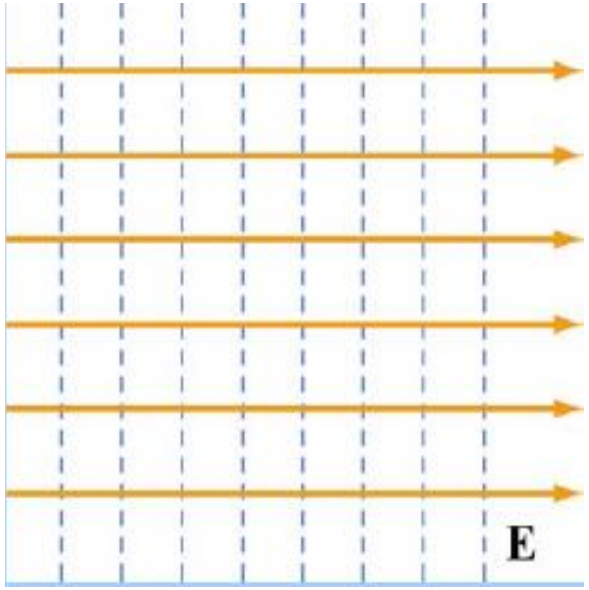
Kurva Equipotensial



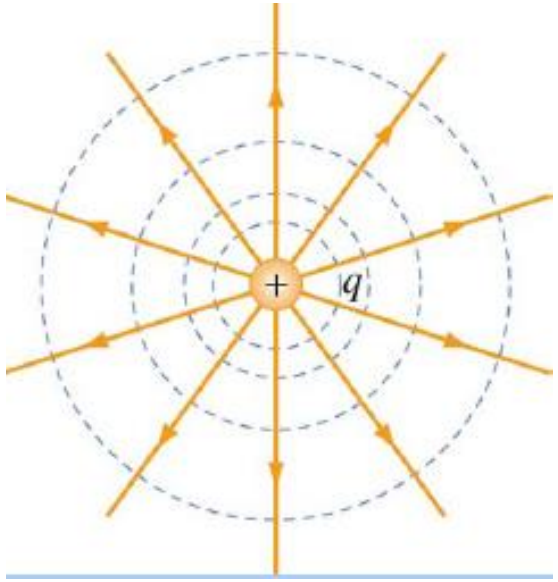
Semua titik pada kurva equipotential memiliki potensial yang sama.
Setiap kurva direpresentasikan oleh V(x,y) = konstan

Arah Medan Listrik E

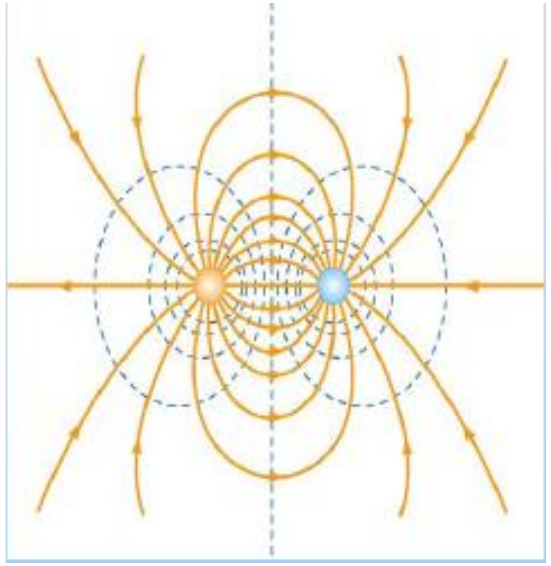
E tegak lurus pada semua kurva equipotential



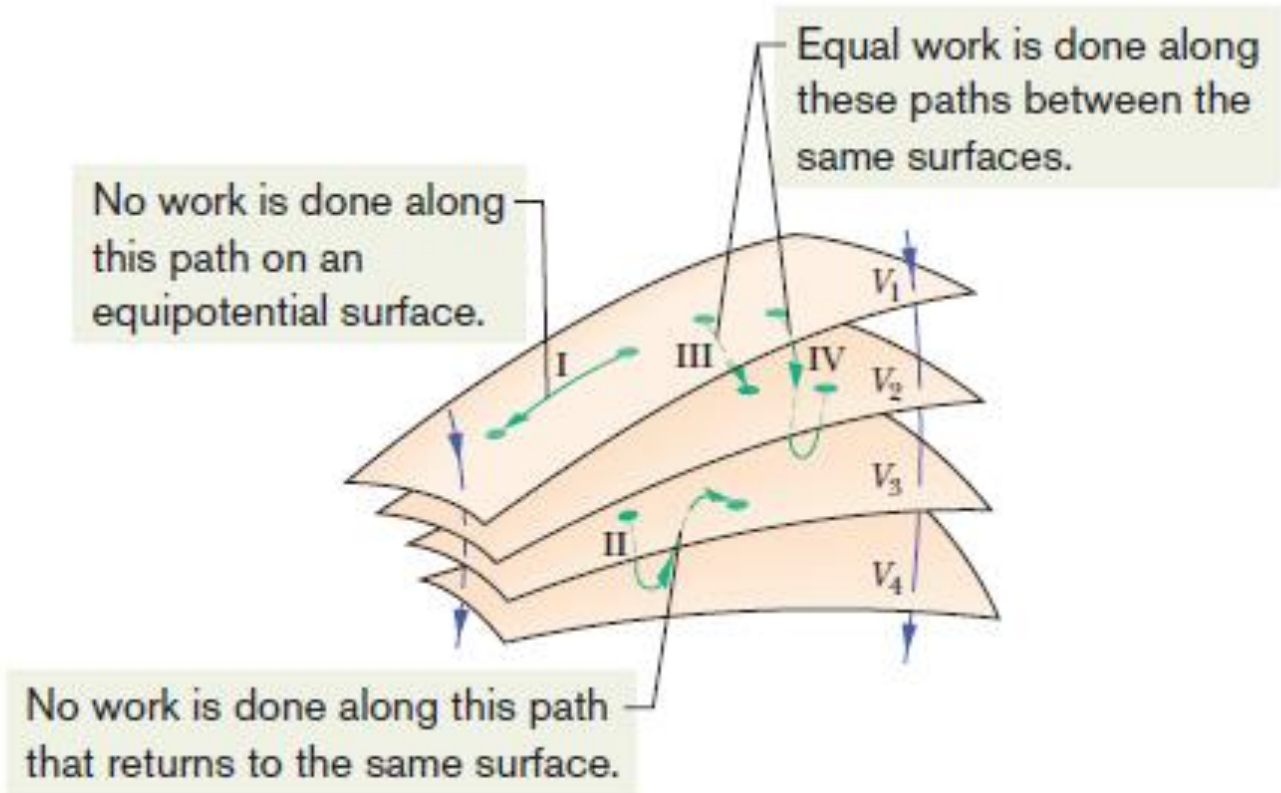
E konstan



Muatan Titik



Dipol Listrik


Equal work is done along these paths between the same surfaces.
No work is done along this path on an equipotential surface.
I
III
IV
II
V_1
V_2
V_3
V_4
No work is done along this path that returns to the same surface.

Sifat-sifat Equipotensial

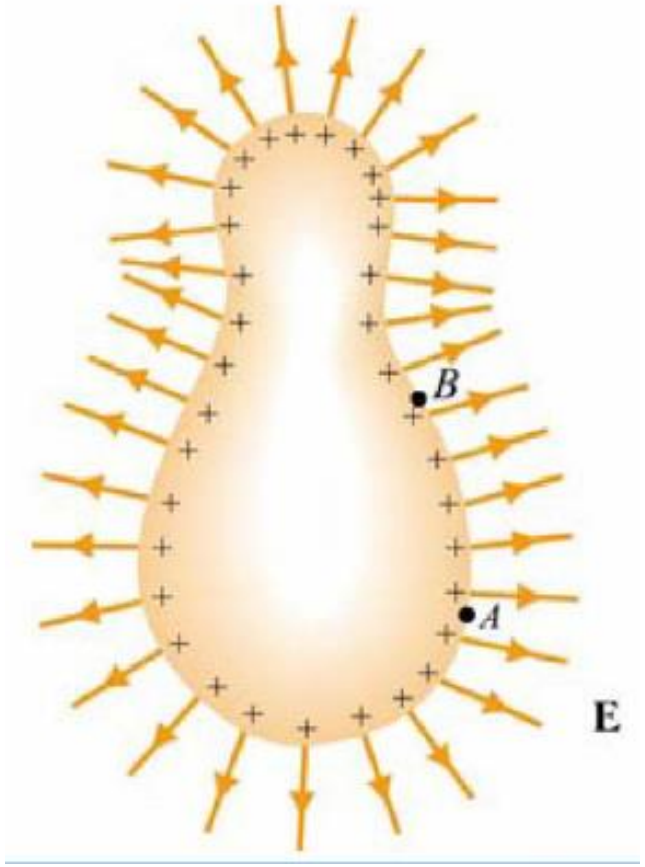
- Garis medan listrik E berarah dari potensial tinggi ke potensial rendah
- Garis medan listrik E tegak lurus dengan equipotensial
  - Tidak punya komponen sepanjang equipotential
  - Tidak ada usaha selama bergerak sepanjang equipotensial

Konduktor dalam Kesetimbangan

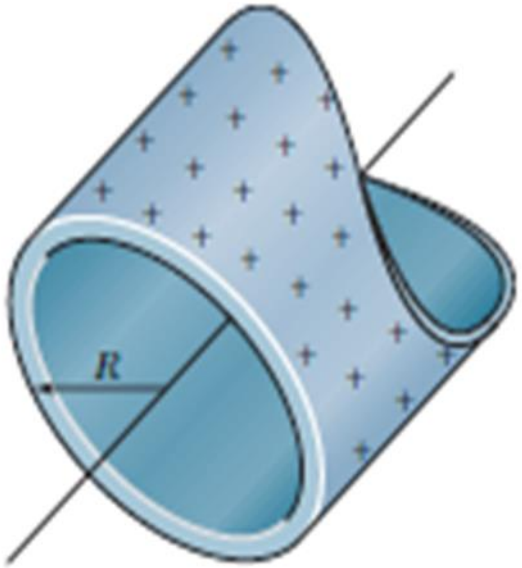
Konduktor adalah objek equipotensial:

1) E = 0 di dalam
2) Muatan neto di dalam 0
3) E tegak lurus permukaan
4) Kelebihan muatan pada permukaan

$$E = \sigma / \varepsilon_0$$



Contoh : Tentukan Potensial di diluar dan didalam konduktor berbentuk kulit selinder dengan rapat muatan panjang konstan λ



Tugas no 1

88 Two charges $q = +2.0\,\mu C$ are fixed a distance $d = 2.0\,cm$ apart (Fig. 24-64). (a) With $V = 0$ at infinity, what is the electric potential at point C? (b) You bring a third charge $q = +2.0\,\mu C$ from infinity to C. How much work must you do? (c) What is the potential energy U of the three-charge configuration when the third charge is in place?

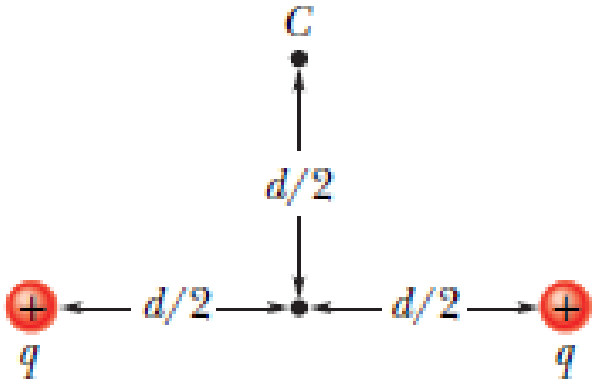
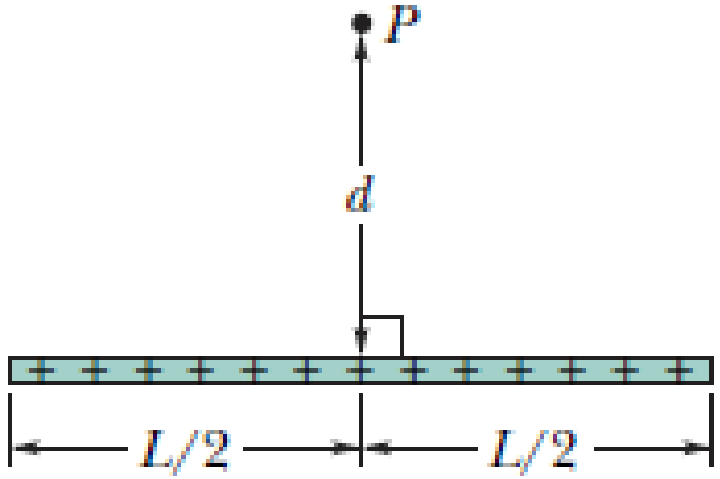


Fig. 24-64 Problem 88.

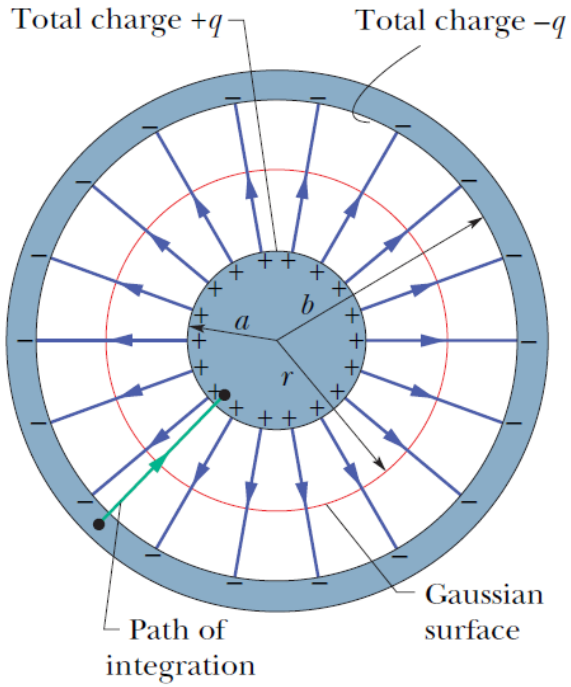
Tugas No 2

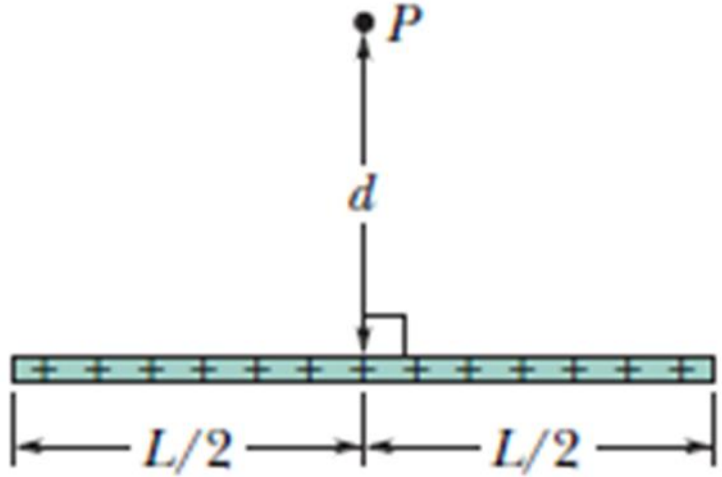
- Tentukan potensial di titik P dari distribusi muatan berbentuk batang seperti pada gambar, rapat muatan panjang konstan λ

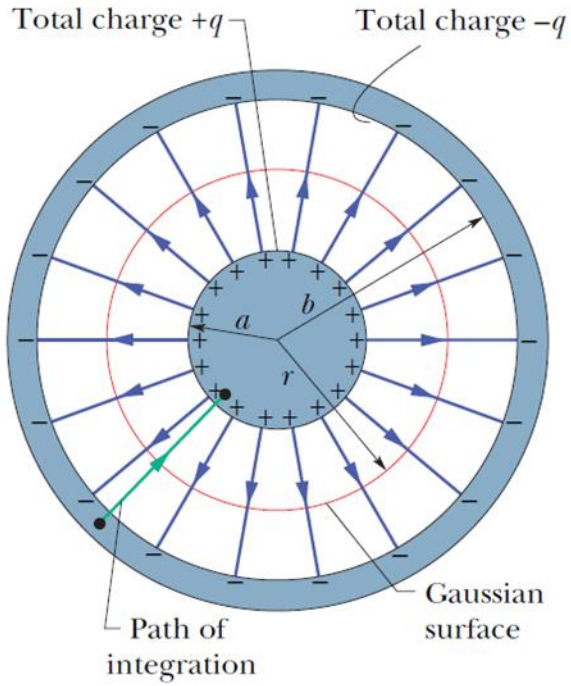


Tugas No 3

Dengan mengunakan hukum gauss tentukan medan listrik pada selinder (a<r<b) konduktor panJang L sepert ipada gambar, dan tentukanlah beda potensial diantara a dan b !




P
d
L/2
L/2


Total charge +q
Total charge −q
a
b
r
Gaussian surface
Path of integration